

RESEARCH ARTICLE

Energy-Efficient Driving Waveform Design for E-Paper Display

Jinhui Lai^{1,2}  | Qudeng Xiangba³ | Yuxiang Fu^{1,2} | Mingrui Cao^{1,2} | Nian He⁴ | Yiwen Xu³

¹Maynooth International Engineering College, Fuzhou University, Fuzhou, China | ²Maynooth International Engineering College, National University of Ireland Maynooth, Maynooth, County Kildare, Ireland | ³College of Physics and Information Engineering, Fuzhou University, Fuzhou, China | ⁴Zhicheng College, Fuzhou University, Fuzhou, China

Correspondence: Nian He (02121003@fdzcx.edu.cn)

Received: 15 July 2025 | **Revised:** 18 December 2025 | **Accepted:** 27 January 2026

Keywords: E3DW | energy-efficient driving | square-wave driving | temperature compensation

ABSTRACT

Color electronic paper (E-paper) has been widely adopted in practical applications across various aspects of daily life. However, conventional driving schemes exhibit high-energy consumption, slow refresh speed, and poor display stability, which restrict their applicability in portable and internet of things (IoT) scenarios. In this paper, we propose an energy-efficient E-paper driving waveform (E3DW) method to improve energy efficiency under varying environmental conditions. A temperature compensation mechanism and a dynamic voltage adjustment strategy are jointly integrated to adapt the waveform in real time based on ambient temperature and image content. The proposed method significantly reduces energy consumption by up to 35.7%, suppresses ghosting artifacts, and enhances display stability. Experimental evaluations on a practical platform verify its superiority over mainstream schemes.

1 | Introduction

Color E-paper simulates the display characteristics of conventional paper, offering significant advantages including ultra-low carbon emissions, thin form factors, and minimal power consumption [1]. It provides excellent readability under strong ambient illumination [2] and supports rich color rendering [3], substantially enhancing the visual experience. Consequently, color E-paper has found widespread adoption in applications such as e-book readers, electronic shelf labels, smart cards, and wearable devices [4]. However, compared to its monochrome counterpart, color E-paper presents greater driving challenges, demanding more complex waveform designs, higher driving voltages, and longer refresh durations. These limitations lead to increased power consumption and hinder large-scale deployment. Therefore, optimizing driving waveforms in conjunction with system-level energy-saving strategies is crucial to overcome these technical bottlenecks and extend the practical applicability of color E-paper. Recent research has primarily

focused on three domains: waveform optimization, temperature compensation, and dynamic voltage control [5]. For waveform optimization: Lookup table (LUT)-based methods [6] enable dynamic waveform selection for grayscale transitions but impose substantial memory overhead from stored patterns. Four-stage schemes [7] reduce ghosting by sequentially executing reset, black-white preset, color setting, and hold phases, but at the cost of increased refresh latency. Trapezoidal waveforms [8] achieve approximately 30% power savings via edge-slope adjustments but require high-precision circuitry, increasing hardware complexity. Hybrid approaches that combine square waves and PWM [9] improve energy efficiency while increasing system complexity. For temperature compensation, closed-loop feedback control [10, 11] ensures display stability across -10°C to 50°C but introduces additional hardware overhead. Model-based methods [12] that correlate temperature with microcapsule response time demonstrate accuracy in controlled settings but exhibit sensitivity to external disturbances. In dynamic voltage control, content-adaptive strategies [13] adjust voltage based on image

contrast, achieving up to 25% energy savings. Frame-difference techniques [14] further reduce consumption by applying voltage only to changed pixels. Region-aware strategies [10] partition display content and apply differentiated voltages and waveforms to balance energy efficiency and visual quality. Despite these advancements, existing approaches exhibit the following: single-objective focus, lack of system-level coordination, limited field validation, and insufficient research on square-wave driving optimization. To address these limitations, this work proposes the energy-efficient E-paper driving waveform (E3DW) method, validated through theoretical modeling and experiments. The main contributions are as follows:

- A modular implementation framework is constructed for an energy-efficient driving system of color E-paper, which supports the high-efficiency driving process of color E-paper.
- An optimized square-wave waveform with dynamically adjustable parameters is designed, enabling the driving strategy to be flexibly adjusted according to different display contents

2 | Principle and Analysis

2.1 | Framework of E3DW

To implement the proposed E3DW, this study constructs the framework illustrated in Figure 1

Figure 1 illustrates the four core modules and their essential functions within the proposed framework, as detailed below.

1. **Square-Wave Driving Module:** Display content data are input to this module, which generates an optimized driving waveform tailored to the characteristics of the E-paper display.
2. **Temperature Compensation Module:** This module processes data from temperature sensors and adjusts the driving waveform's voltage amplitude, duty cycle, and timing parameters based on the measured environmental temperature.
3. **Dynamic Voltage Adjustment Module:** This module receives square-wave signals, matches the optimal driving voltage according to the displayed content of the signals, and then inputs the driving voltage into the driving timing control module.

4. **Driving Sequence Control Module:** This module synchronizes the temperature-compensated square-wave signal with the dynamically adjusted driving voltage. It then outputs the final driving signals to the source and gate drivers according to a predefined timing sequence, thereby refreshing the E-paper display.

2.2 | Theoretical Analysis of Optimization Strategies for Dynamically Tunable Driving Parameters

Through experimental characterization and theoretical analysis, the approximate relationship between the displacement of pigment particles within the microcapsules and time can be described by the model in Equation (1):

$$x(t) = x_{\max}(1 - e^{-t/\tau}) \quad (1)$$

where $x(t)$ denotes the particle displacement at time, $x_{\max}$ is the maximum displacement distance, and τ is the time constant of particle motion. This model indicates the intrinsic nonlinear dynamic behavior of particles. During the early stage of driving, the particles exhibit relatively high velocity, and the displacement increases almost linearly with time. As the particles approach their final position, the velocity gradually decreases and eventually converges to $x_{\max}$. Consequently, continuing to apply a high-amplitude stimulus in this saturation region yields minimal additional displacement while unnecessarily increasing energy consumption, thereby reducing driving efficiency. The relationship between particle velocity and applied voltage can be approximated by Equation (2):

$$v = \frac{QE}{6\eta\pi r} \quad (2)$$

where v is the particle velocity, Q is the particle charge, E is the electric field strength, η is the viscosity of the fluid, and r is the particle radius. Under the assumption that the system operates within the linear electrophoretic regime and no nonlinear effects such as overdriving, particle aggregation, or dielectric breakdown occur, the electric field strength is proportional to the applied voltage when the electrode spacing is fixed. Therefore, higher driving voltages generally lead to faster particle migration.

Based on the dynamics described by Equations (1) and (2), particle motion can be divided into two phases: a rapid-response

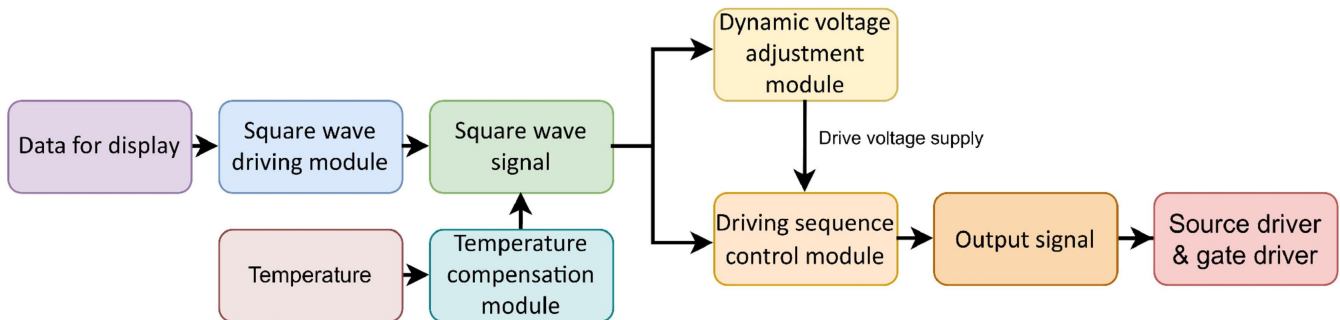


FIGURE 1 | E3DW driving system modular framework schematic diagram.

phase and a saturation phase. Accordingly, an optimization strategy is proposed wherein driving parameters are dynamically adjusted according to the motion phase. Higher voltage amplitudes or wider pulse widths are applied during the initial phase to accelerate early displacement, whereas reduced voltage levels or narrower pulse widths are used in the mid-to-late stages to maintain stable motion and achieve precise positioning, thereby avoiding unnecessary energy expenditure. This phase-aware driving strategy enhances overall energy efficiency.

In this work, the optimization is not expressed as a single mathematical cost function. Instead, driving parameters are co-optimized under the following practical constraints: (1) minimizing refresh energy consumption, (2) ensuring display quality remains within an empirically defined acceptable range, and (3) maintaining operational stability across temperature variations. Parameter selection thus becomes a constrained optimization problem, where energy reduction is prioritized while limiting the degradation of contrast and color saturation to experimentally tolerable levels.

2.3 | Optimized Square-Wave Driving Parameters

In electrophoretic display driving, square-wave driving is widely adopted as a fundamental method due to its circuit simplicity [15]. Conventional square-wave waveforms, however, typically employ fixed voltage amplitudes, a single frequency, and constant duty cycles. When applied to color electrophoretic displays—which demand higher performance—these static parameters lack dynamic adaptability. This often results in reduced energy efficiency, ghosting artifacts, and degraded response speed and image quality under low-temperature conditions or when rendering complex content [16–18].

The optimization objective of this work is to dynamically adjust the driving voltage, frequency, and duty cycle of the square wave. By adapting these parameters to ambient temperature and display content variations, overall system energy efficiency can be enhanced.

To achieve the objectives of E3DW, several specialized waveform strategies were developed (Figure 2), each with tailored parameter optimizations.

Three-stage transition (Waveform 1) is divided into reset, driving, and stabilization phases. The reset phase applies specific pulses to clear residual pixel charges, suppressing memory effects and providing a clean state for grayscale writing to mitigate ghosting. The driving phase uses energy-optimized pulses for rapid grayscale transitions. The stabilization phase employs short pulses or zero-voltage holding to enhance microcapsule state stability, improving final image clarity.

Fine-grained driving (Waveform 2) is designed for rendering intermediate tones and color blending. Using narrower pulse widths and lower voltages, it enables precise particle displacement control for accurate and delicate grayscale reproduction.

Particle activation (Waveform 3) rapidly alternates voltage polarity to overcome viscous forces between pigment particles and

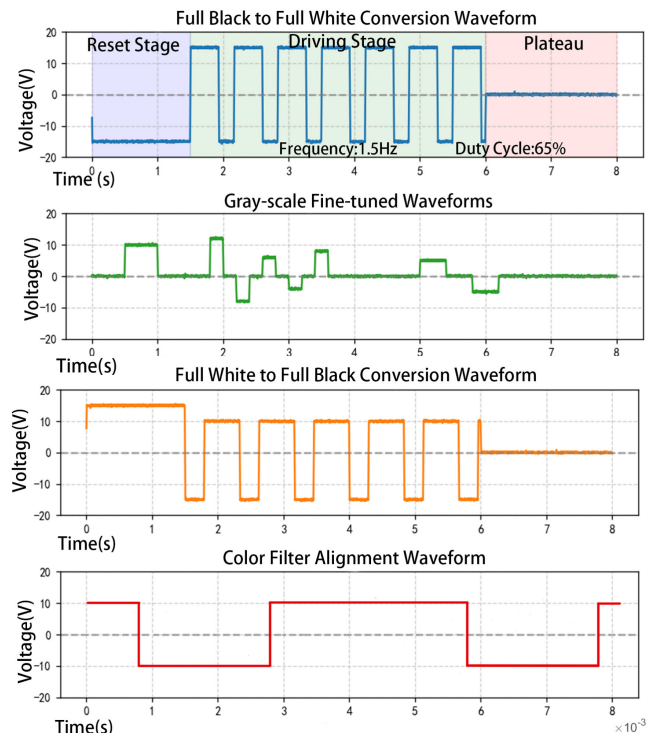


FIGURE 2 | Optimized square-wave driving waveform.

microcapsule walls. This brings particles into an active state prior to precise color mixing, which helps reduce ghosting and improve color purity.

Color filter alignment (Waveform 4) is tailored for alignment with the color filter layer, which imposes strict light-transmission requirements. Combining activation pulses with duty-cycle modulation, this waveform enhances particle activation at the physical level to improve brightness while enabling fine positional control for accurate color alignment, ensuring high-quality display performance.

The E3DW implementation incorporates four key parameter configurations:

1. **Multilevel Voltage Configuration:** This paper designs a three-level driving voltage scheme. The low voltage level of ± 10 V is applied to high-contrast pure text and black-and-white line patterns, optimizing performance for simple display scenarios. The medium voltage level of ± 12.5 V targets simple color blocks and icons, while the high voltage level of ± 15 V is used for full-color photographs and complex grayscale images. By selecting the minimum effective driving voltage according to actual grayscale transition requirements, power consumption can be effectively reduced. Moreover, this voltage strategy provides a necessary foundation for the subsequent dynamic voltage adjustment mechanism.
2. **Low-Frequency Operation:** A 1 to 5-Hz square-wave frequency range reduces charge-discharge cycles, lowering energy consumption and mitigating charge accumulation in the microcapsule medium.

3. **Poly-Phase Configuration:** 120° phase-shifted waveforms across electrophoretic channels attenuate current ripple, enhancing system stability and power efficiency.
4. **Duty Cycle Configuration:** The duty cycle is adjusted within 30%–70% based on grayscale transition direction: higher (60%–70%) for dark-to-light transitions to accelerate particle ascent, lower (30%–40%) for light-to-dark to prevent overdriving, and midrange (45%–55%) for small grayscale changes to balance energy and accuracy. This strategy enhances both driving efficiency and grayscale control.

2.4 | Driving Framework Based on Other Techniques

Temperature Compensation Module: This module implements the strategy reported in [6], utilizing a combination of LUTs and linear interpolation. Optimal driving parameters (voltage, timing, frequency, duty cycle) are prestored at key calibration temperatures (e.g., -20°C , 0°C , 25°C , and 60°C). During operation, the system reads the current temperature via sensors. If the temperature falls between two calibration points, the corresponding parameters are calculated in real-time through linear interpolation. This enables adaptive adjustment: Driving voltage and pulse duration are increased under low temperatures to counteract slower particle mobility, while they are decreased under high temperatures to prevent over-driving and save energy. Similarly, lower frequencies and higher duty cycles are applied in cold environments, with the opposite adjustments made in warm conditions.

Dynamic Voltage Module: According to [19], a content-aware dynamic voltage strategy works in concert with the core waveform. This module analyzes the grayscale histogram of each input frame to classify content complexity. Key extracted statistical features include the following: (1) the number and location of histogram peaks (e.g., isolated peaks indicate simple text/graphics), (2) the entropy of pixel distribution (low entropy suggests simple content, high entropy indicates complexity), and (3) the standard deviation of grayscale values (reflecting overall contrast). Based on this classification, the driving voltage is scaled adaptively—lower voltages for simple content (e.g., text) and higher voltages for complex images (e.g.,

photographs)—enabling efficient, content-proportional energy management.

Refresh Strategy: Following the method described in Reference [20], the system employs a frame-difference-based refresh strategy. The system continuously compares a new input frame with the currently displayed frame to compute a pixel-wise difference map. If the difference map is empty (i.e., no pixel has changed), the refresh cycle is skipped, avoiding all driving power consumption and eliminating potential ghosting [21]. If changes are detected, only the affected pixel regions are updated via a partial refresh. This selective updating significantly reduces energy consumption. The complete workflow is illustrated in Figure 3.

3 | Experiments

3.1 | Experimental setup

Evaluation: The proposed method was compared against conventional multistage waveform driving [15], improved PWM approach, and trapezoidal waveform method [8]. Evaluation metrics included the following:

- Average refresh energy consumption
- Contrast ratio (reflectance ratio between black and white regions)
- Color saturation (CIELAB color space)
- Refresh time
- Temperature stability (performance variation across temperatures)

Implementation Details: An experimental platform was constructed comprising: (1) an ARM Cortex-M4 microcontroller for system control; (2) a driver board integrating waveform generation, high-voltage driving, and timing control circuits; (3) a temperature chamber adjustable from -20°C to 60°C for environmental simulation; and (4) a measurement system with a power analyzer, oscilloscope, and optical instruments to quantify energy consumption and display performance. The E Ink SD1452-NCB (EC133UJ1) color E-paper panel, based on Carta technology, features a 13.3-inch display, 2200×1650 resolution, and grayscale/

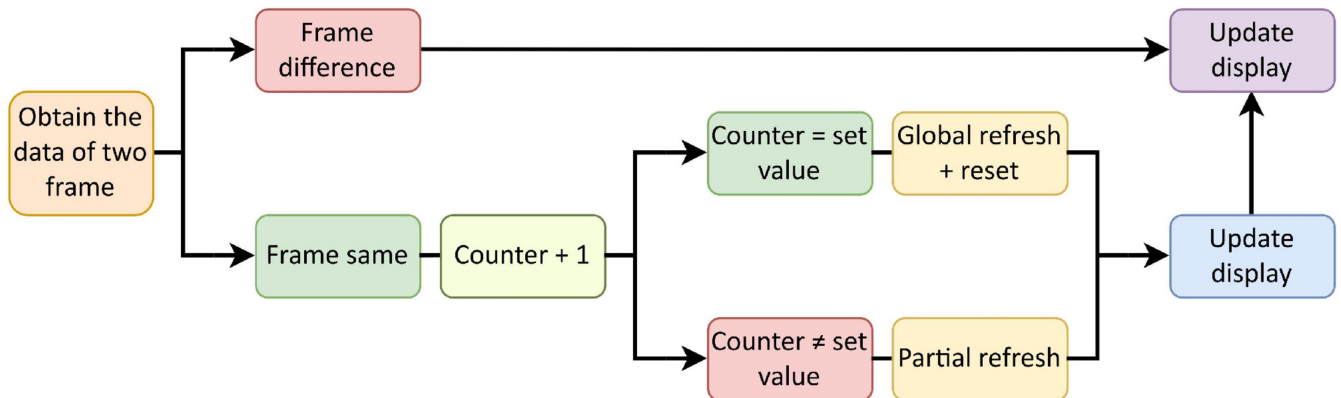


FIGURE 3 | Refresh strategy workflow.

color rendering support. Figure 4 shows the experimental platform, enabling comprehensive validation of the proposed E3DW scheme and its cooperative energy-efficient strategy under practical conditions.

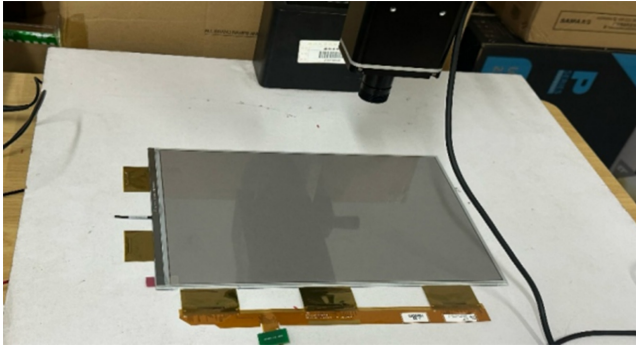


FIGURE 4 | Experimental platform.

3.2 | Experimental Result and Analysis

Temperature Stability Evaluation: Full-screen color refresh tests were conducted on a 6-inch color e-panel across -20°C to 50°C . Figure 5 demonstrates E3DW's energy-efficient operation and superior thermal stability, exhibiting a flatter consumption curve versus conventional driving's significant low-temperature energy surge, indicating enhanced subzero adaptability.

Energy Consumption Evaluation: Comparative energy tests were performed on four display content types using conventional driving and E3DW method. As shown in Figure 6, E3DW achieves consistent energy reductions across content types, yielding approximately 35.7% average energy savings versus conventional driving.

Refresh Time Evaluation: As shown in Figure 7, refresh times of the E3DW method were measured against conventional

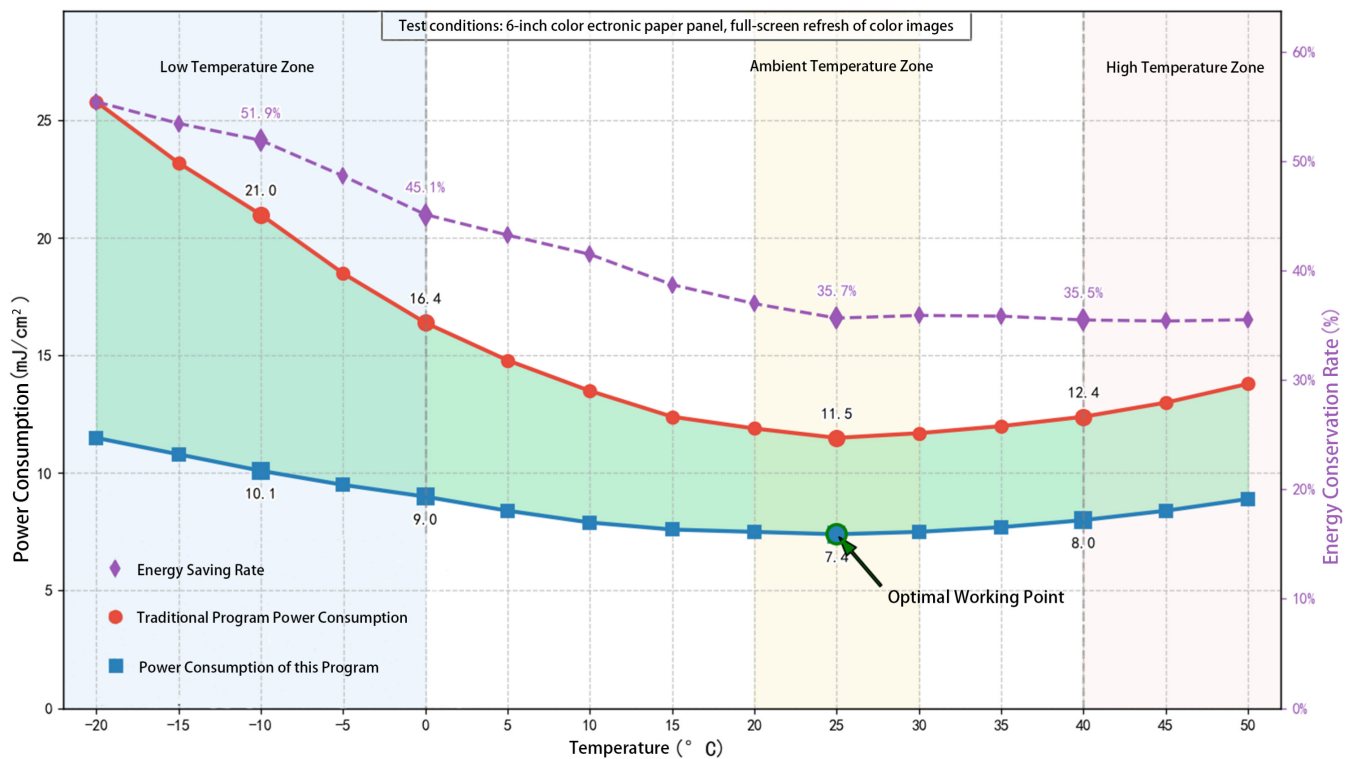


FIGURE 5 | Comparative analysis of power consumption between the conventional driving scheme and the proposed method across different temperatures.

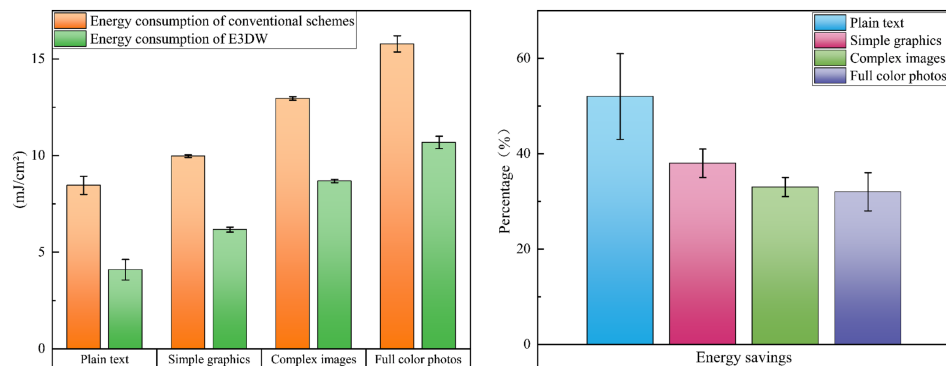


FIGURE 6 | Comparison of energy consumption of different display contents (left) and energy savings of the E3DW scheme (right).

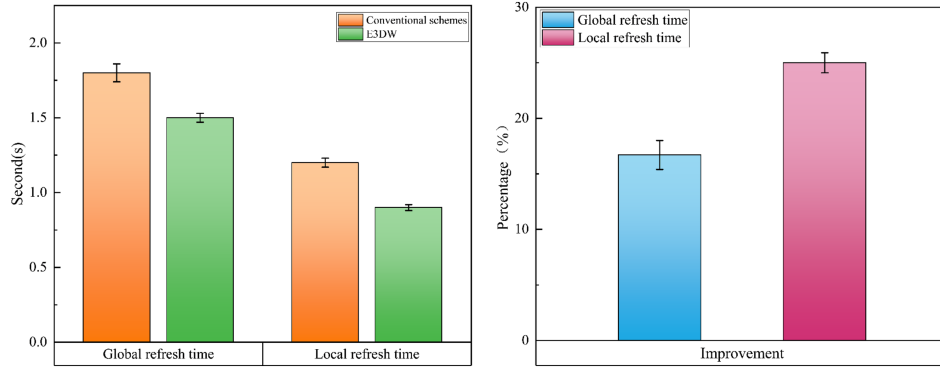


FIGURE 7 | Refresh time comparison (Left) and improvement of the E3DW scheme (Right).

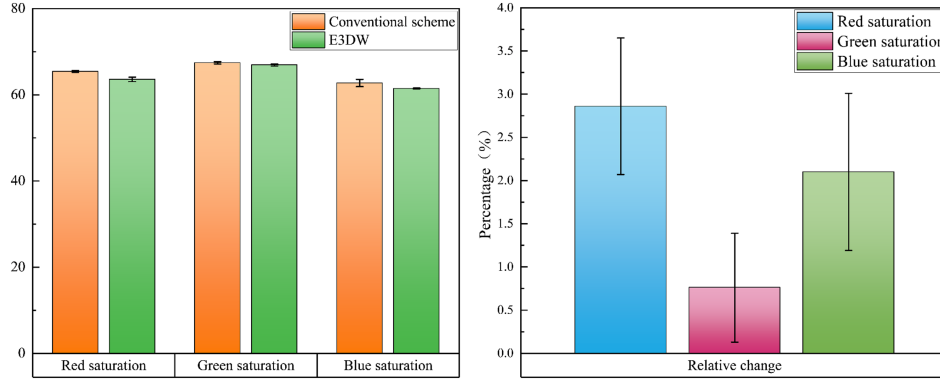


FIGURE 8 | Display quality comparison (left) and relative change (right) (conventional scheme vs. E3DW scheme).

TABLE 1 | Comparison between E3DW and mainstream driving schemes.

Evaluation indicators	Conventional multiphase waveform	Improved PWM scheme	Trapezoidal wave scheme	E3DW
Energy consumption (mJ/cm ²)	11.47 ± 0.14	9.17 ± 0.23	8.52 ± 0.10	7.46 ± 0.06
Black and white contrast	(10.1 ± 0.3):1	(10.0 ± 0.1):1	(10.3 ± 0.2):1	(9.9 ± 0.1):1
Average color saturation	65.10 ± 0.30	64.77 ± 0.11	65.30 ± 0.26	64.87 ± 0.24
Refresh time (s)	1.79 ± 0.03	1.56 ± 0.21	1.73 ± 0.07	1.25 ± 0.11
Implementation complexity	High	Middle	High	Low
Temperature adaptability	Middle	Middle	Middle	High
Cost efficiency	High	Middle	High	Low

multistage waveform driving. Results indicate significant improvements in both global/local refresh times with E3DW. The accelerated process improves user experience while shortening active driving duration, further reducing overall system power consumption.

Due to the performance limitations of the general-purpose control hardware used in this experiment, the measured refresh time is longer than that of commercial products. If matched with a commercial-grade dedicated driver chip, the refresh time of the E3DW scheme can be further optimized to 0.85 ± 0.04 s, which is close to the response level of commercial color E-paper of the same specification.

Display Quality Evaluation: As presented in Figure 8, E3DW was evaluated against four key display metrics. Compared to conventional driving, it exhibits a modest 4% reduction in display quality, accompanied by a 35.7% reduction in energy consumption, indicating a clear energy-quality trade-off. This confirms E3DW's optimal balance between energy efficiency and visual performance.

Comprehensive Comparison With Mainstream Driving Schemes: Table 1 compares E3DW with established driving schemes, highlighting significant advantages across five key metrics: power consumption, refresh speed, implementation complexity, temperature adaptability, and cost efficiency.



FIGURE 9 | Conventional square-wave image generation (left) and E3DW scheme (right).

Implementation complexity and cost efficiency are assessed qualitatively according to hardware requirements, waveform generation precision, and control logic complexity.

Display Performance Comparison: As shown in Figure 9, conventional method exhibits inconsistent grayscale transitions, color shifts, and visible ghosting, compromising visual clarity. Conversely, E3DW demonstrates enhanced grayscale accuracy, improved color fidelity, and effective ghosting suppression, delivering superior visual stability and naturalness.

4 | Conclusions

This work proposes an energy-efficient driving method for color E-paper, based on an optimized square-wave approach integrated with temperature compensation and dynamic voltage adjustment. Experimental results demonstrate significant improvements in power consumption control, thermal adaptability, and ghosting suppression. While limitations persist in achieving premium-grade display quality, the proposed method achieves an optimal balance among performance, cost, and energy efficiency. Future work will investigate machine-learning-based adaptive algorithms, application-specific optimizations, hardware acceleration, and chip-level integration to advance practical deployment.

Acknowledgments

The authors would like to thank Prof. Zhao for valuable guidance and insightful discussions throughout this work. We also acknowledge Mr. Zhong for his assistance in experimental setup and data analysis. This work was supported by the Natural Science Foundation of Fujian Province, China (2024J011029) and the National Key Research and Development Program of China (2022YFB3603704).

Data Availability Statement

Research data are not shared.

References

1. Y. Lu, Y. Gu, L. Zhou, and H. Yang, "A New Waveform to Solve the Ghosting of BWR EPD," *Micromachines* 16, no. 4 (2023), <https://doi.org/10.3390/mi16040420>.
2. J. Liu, G. Liu, Y. Liu, et al., "High Contrast Ratio Electrophoretic Displays Based on High Structural Stability and High Transmittance Microcup," *Optics Express* 31, no. 23 (2023): 37523–37536.
3. X. M. Le, Q. S. Tran, T. T. Vu, T. V. Vo, L. H. Trinh, and F. Ferrero, "Research and Design a Low Power Electronic Shelf Label Based on E-Paper Display and LoRa Technology," in *In2023 International Conference on Advanced Technologies for Communications (ATC)* (IEEE, 2023), 356–360.
4. M. Nose and T. Yoshihara, "Quantification of Image Quality for Color Electronic Paper," *Journal of the Society for Information Display* 20, no. 11 (2012): 624–631.
5. D. Kang, X. Zhao, X. Wei, H. Wu, H. Zhang, and T. Zhao, "Effective E-Paper Driving Waveform Design Based on Dynamic Programming," *Journal of the Society for Information Display* 33, no. 12 (2025): 1114–1122.
6. Y. Yang, Y. Xie, J. Yuan, Y. Cui, Q. Cheng, and L. Shi, "A New Waveform to Solve the Ghosting of BWR EPD," *Micromachines* 16, no. 4 (2025): 420.
7. G. Yang, L. Liu, Z. Zheng, et al., "A Portable Driving System for Highresolution Active Matrix Electrowetting Display Based on FPGA," *Journal of the Society for Information Display* 28, no. 3 (2020): 287–296.
8. K. Taniguchi and H. Irie, "Trapezoidal Modulating Signal for Three-Phase PWM Inverter," *IEEE Transactions on Industrial Electronics* 23, no. 2 (2007): 193–200.
9. SJ Telfer, WT Chen, MD McCreary. 671: Fullcolor Electrophoretic Displays With Improved Update Time. *InSID Symposium Digest of Technical Papers* 2021 (Vol. 52, 1, pp. 995–998).

10. Y. Zhuang, W. Liu, T. Xie, et al., "Temperature Compensated Optical Fiber Current Sensing System Based on a Two-Tap Microwave Photonic Filter," *Optics Communication* 9 (2025): 132106.
11. Y. Xu, D. Liu, Z. Lin, T. Zhao, and N. He, "MSFEN-AM: A Non-Intrusive Load Identification Method for Power Saving," in *In2023 IEEE 11th International Conference on Information, Communication and Networks (ICICN)* (IEEE, 2023), 675–680.
12. W. He, Z. Yi, S. Shen, et al., "Driving Waveform Design of Electrophoretic Display Based on Optimized Particle Activation for a Rapid Response Speed," *Micromachines* 11, no. 5 (2020): 498.
13. G. Y. Wei and M. Horowitz, "A Fully Digital, Energy-Efficient, Adaptive Power-Supply Regulator," *IEEE Journal of Solid-State Circuits* 34, no. 4 (1999): 520–528.
14. B.-R. Yang, P.-Y. Cheng, C.-M. Hung, C.-A. Wei, Y.-J. Hsieh Method for Displaying an Image on an Electrophoretic Display. US Patent 10,229,641 B2. 2020 Mar 19. <https://patents.google.com/patent/US10229641B2/>.
15. S. Lin, J. Zhang, J. Wei, et al., "Low-Power Driving Waveform Design for Improving the Display Effect of Electrophoretic Electronic Paper," *Micromachines* 15, no. 9 (2024): 1076.
16. Gu Y, Zhu S, Qiu Z, Fan Q, Shi J, Chen W, Peng L, Qin Z, Deng S, Yang BR. 252: Capacitor-Based Driving Scheme of Electrophoretic E-Paper Display for Future Self-Powered Applications. *InSID Symposium Digest of Technical Papers* 2024 (Vol. 55, 1, pp. 314–317).
17. Q. An, P. Bai, A. Henzen, B. Tang, and G. Zhou, "16 Gray-Scale Driving Schema Based on DC-Balance Principle of Electrophoretic Display," in *In2016 IEEE International Conference on Signal and Image Processing (ICSIP)* (IEEE, 2016), 712–716.
18. J. Shi, Q. Fan, S. Zhu, et al., "Dual-Mode Flexible Electrophoretic E-Paper With Integration of Alternating Current Electroluminescent Technology for Ubiquitous Ambient Light Applications," *Advanced Functional Materials* 34, no. 51 (2024): 2410139.
19. J. M. Kim, K. Kim, and S. W. Lee, "Multilevel Driving Waveform for Electrophoretic Displays to Improve Grey Levels and Response Characteristics," *Electronics Letters* 50, no. 25 (2014): 1925–1927.
20. B. R. Yang, W. J. Hu, Z. Zeng, et al., "Understanding the Mechanisms of Electronic Ink Operation," *Journal of the Society for Information Display* 29, no. 1 (2021): 38–46.
21. W. C. Kao, C. H. Liu, S. C. Liou, J. C. Tsai, and G. H. Hou, "Towards Video Display on Electronic Papers," *Journal of Display Technology* 12, no. 2 (2016): 129–135.